

TREE INDEPENDENCE SEQUENCES ARE UNIMODAL THROUGH ORDER 31

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ABSTRACT. We give an exact exhaustive verification that every tree on exactly 31 vertices has a unimodal independent-set sequence. The computation streams all 40,330,829,030 unlabelled trees from `gentreeg`, evaluates each independence polynomial by integer rooted-tree dynamic programming, and partitions the enumeration into 120 disjoint residue classes. Two fail-closed aggregators agree on the exact census, on two order-independent fingerprints, and on the absence of a unimodality failure. A separately written Python evaluator reconstructs all 159 order-31 trees whose sequences fail log-concavity and confirms that all remain unimodal. An isolated rebuild of the generator plugin reproduces the validation fingerprints. Together with the previously reviewed verification through order 30, this raises the finite no-counterexample frontier for the Alavi–Malde–Schwenk–Erdős conjecture to 31. The conjecture for arbitrary trees remains open.

1. INTRODUCTION

For a finite graph G , let $i_k(G)$ be the number of independent vertex sets of size k . Alavi, Malde, Schwenk, and Erdős asked whether the sequence

$$i_0(T), i_1(T), \dots, i_{\alpha(T)}(T)$$

is unimodal for every tree (and perhaps every forest) [1]. The question is Erdős Problem 993 and remains open in its unrestricted form.

The stronger assertion of log-concavity is false. Exact examples begin at order 26 [3], and infinite or large searched families are now known [2, 7]. These results do not supply a non-unimodal tree. On the computational side, Reynolds reported an exhaustive verification through order 29 [8]; a later independently reviewed computation extended the frontier to order 30 [9]. The purpose of this note is the next exact layer.

Our main result is deliberately finite. It is a computational theorem about all trees of order 31, not an induction step and not evidence promoted to a claim about unbounded order. The full serialized certificate, the frozen sources, two aggregators, and independent reconstruction code accompany the paper.

Date: 2026-08-29.

2. DEFINITIONS AND REDUCTIONS

The independence polynomial of a finite simple graph is

$$I_G(x) = \sum_{k=0}^{\alpha(G)} i_k(G) x^k.$$

A finite sequence $(a_0, \dots, a_q)$ is *weakly unimodal* if some $p \in \{0, \dots, q\}$ satisfies

$$a_0 \leq \dots \leq a_p \geq \dots \geq a_q.$$

Equivalently, there do not exist $k < \ell < m$ with $a_k > a_\ell < a_m$. Indeed, one implication is immediate; for the other, choose an index of a global maximum and observe that a descent before it or an ascent after it would create such a strict triple valley.

Root a tree at r . Let A_T count independent sets not containing r , and let B_T count those containing r . If $T_1, \dots, T_d$ are the child subtrees, then

$$(1) \quad A_T = \prod_{j=1}^d (A_{T_j} + B_{T_j}), \quad B_T = x \prod_{j=1}^d A_{T_j}, \quad I_T = A_T + B_T.$$

The leaf boundary is $A_T = 1$, $B_T = x$. Equation (1) follows by partitioning independent sets according to whether the root is chosen and is valid over $\mathbb{Z}_{\geq 0}[x]$.

For a 31-vertex tree,

$$(2) \quad i_k(T) \leq \binom{31}{k} \leq \binom{31}{15} = 300,540,195.$$

Every intermediate convolution accumulator in (1) counts a class of independent sets in a processed subforest, so the same binomial bound applies. The largest product needed for a log-concavity comparison is therefore at most

$$300,540,195^2 = 90,324,408,810,638,025 < 2^{64} - 1.$$

Thus all decisive arithmetic in the order-31 checker is exact in unsigned 64-bit integers. Hash arithmetic, used only for integrity fingerprints, is intentionally taken modulo 2^{64} .

3. MAIN RESULT

Theorem 3.1. *Every tree on exactly 31 vertices has a weakly unimodal independent-set sequence.*

Proof. For the enumeration, we used `gentreeg` from nauty 2.9.3. Its bundled source identifies the algorithm and code of Li and Ruskey [4] as the basis of the generator; nauty itself is described by McKay and Piperno [5]. We ran all residue classes $0/120, \dots, 119/120$. For every parent array, a plugin reconstructed child lists independently of the vertex order, rejected malformed or disconnected arrays, and evaluated (1) in recursive postorder.

It then tested the coefficient sequence for a strict ascent after the first strict descent, allowing equalities throughout.

Each residue wrote its exception stream and summary first to temporary files with process-specific names and renamed them only after successful completion. Every one of the 120 final summaries reported that the number checked equalled the number generated. Their exact sum was

$$40,330,829,030,$$

which is the number of unlabelled trees on 31 vertices in OEIS A000055 [6]. No residue reported a non-unimodal polynomial.

The production aggregator and a separately written verifier, which imports no project module and uses a different parser, both required all 120 pairs of completed files and rejected any unexpected completed-file name. They agreed on the total above, on zero non-unimodal trees, on the coefficient fingerprint 92f46f1b00c219ad, and on the parent-array fingerprint d53b120ed8c90b52. The second verifier also reconstructed every serialized exceptional polynomial using Python integers. Hence every unlabelled tree of order 31 was checked and passed. Since the independence sequence is invariant under graph isomorphism, the assertion follows for every labelled or unlabelled tree of that order. $\square$

Corollary 3.2. *Every tree on at most 31 vertices has a weakly unimodal independence-set sequence.*

Proof. Theorem 3.1 handles order 31. The independently reviewed prior census handles all orders through 30 [9]. $\square$

Proposition 3.3. *Exactly 159 trees on 31 vertices have non-log-concave independence sequences. Every one of the 159 sequences is unimodal.*

Proof. The production checker serialized the parent array and complete coefficient sequence of every log-concavity failure. The independent verifier required exactly 159 such records, rebuilt each tree and polynomial with arbitrary-precision integers, reproduced a strict log-concavity failure, and separately retested weak unimodality. All 159 reconstructions passed. $\square$

4. CERTIFICATE OR COMPUTATIONAL VERIFICATION

Table 1 records the decisive aggregate. The CPU time is the sum reported by the 120 checker processes; it is not used as evidence of coverage.

Three independent validation layers precede the full census. First, a debug build serialized every polynomial for all 436 unlabelled trees through order 11, and direct enumeration of all vertex subsets reproduced every coefficient. Second, a 24-way partition at order 23 reproduced the exact unsplit count 14,828,074 and both unsplit fingerprints. Third, after the order-31 run, a fresh extraction of the frozen nauty source archive and an isolated rebuild reproduced an order-23 residue count and both fingerprints, while its debug binary again passed the subset audit through order 11.

Quantity	Exact value
Residue classes	120/120
Trees checked and generated	40,330,829,030
Non-unimodal sequences	0
Non-log-concave sequences	159
Coefficient fingerprint mod 2^{64}	92f46f1b00c219ad
Parent-array fingerprint mod 2^{64}	d53b120ed8c90b52
Checker CPU seconds	56,320.05

TABLE 1. Exact order-31 aggregate.

The frozen principal SHA-256 values are listed below. In order, they bind the checker source, the checker binary, and the nauty archive.

```
38fad81aaae42ab4fe4d764abd03482535b54483aa308b9b6ce924d8edd4db36
93a463c656872a4e977390fcae8110cbcd558570bf06a8b78e628dad7752c00e
9fc4edae04f88a0f5883985be3b39cf7f898fd6cc96e96b9ee25452743cc1b5b
```

The full manifest binds every released source, log, and verifier.

The core reproduction commands are

```
bash experiments/run_tree_sweep_all.sh 31 120 8 \
  results/order31_sweep
python3 experiments/aggregate_tree_sweep.py \
  --order 31 --modulus 120 --expected 40330829030 \
  --results-dir results/order31_sweep \
  --project-dir .
python3 audit/order31_independent_aggregate.py
sh audit/rebuild_order31_checker.sh
```

5. EQUALITY CASES AND LIMITATIONS

The theorem uses weak unimodality, so central or other plateau equalities are allowed. The checker tests exactly this convention. The proposition about log-concavity is a by-product and is not used to prove the theorem.

The computation establishes no reduction from order $n > 31$ to order at most 31. It therefore cannot prove the original universal conjecture, and it does not address a direct exhaustive census of forests of order 31. Negative structured searches performed during discovery are likewise absent from the proof. The exact endpoint is the finite frontier in Corollary 3.2.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

All decisive counts use integer arithmetic. The release contains the frozen nauty archive, plugin source, and orchestration scripts. It also contains 240 completed chunk files, two aggregate records, the independent Python verifier, the isolated-rebuild audit, and a SHA-256 manifest. No theorem prover or proof assistant was used.

OpenAI Codex assisted with literature searches, program and manuscript drafting, exact experiment orchestration, and adversarial auditing. The finite theorem rests on the serialized enumeration and independently reproducible verifiers described above, not on an unevaluated model output.

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